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Biochemical study of the lipid profile in *Channa punctatus* with water and soil quality

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Abstract- The present analysis explores the lipid profile of *Channa punctatus* (spotted snakehead) in relation to soil quality and dissolved oxygen contents in different aquatic habitats. The research looks at how such environmental conditions, including soil type and exposure to oxygen, affect the lipid metabolism of this animal. Lipid was determined in various fish tissues from places with differing soil and oxygen conditions. The data are intended to provide information for the comprehension of inter-relationship between ecological factors and lipid metabolism, leading to knowledge on the nutritional quality and adaptive physiology in *Channa punctatus*. The results offer valuable information for the evaluation of aquatic ecosystem health and development FMPs.

Keywords: Biochemical Evaluation, Lipid Profile, *Channa punctatus*, Soil Quality, Dissolved Oxygen

INTRODUCTION

The biochemistry of fish lipids also is of interest in order to evaluate their value what respect for nutritional, and body physiological healthy. Lipids also play an essential role in energy storage, survival and cell membrane integrity as well as other metabolic roles.[1] *Channa punctatus*, a fresh water species which is widely distributed across Bihar plays significant role in contributing to local diet and valuable source of protein and other nutrients.[2]

The lipid profile of this fish is influenced by the habitat quality, mainly because of the nature of the soil and dissolved oxygen. The agricultural runoff and pollution in water alter the conditions of the soil and the water, which influences the lipid metabolism of the aquatic organisms.[3] This might affect the nutritional values and health status of the fish. A study was conducted to analyze the lipid profile of *Channa punctatus* collected from various aquatic habitats across Bihar to examine how environmental factors like soil quality and dissolved oxygen affect the biochemical composition. Understanding these relationships can provide insights into the fish's adaptability and nutritional significance in changing environmental conditions.

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MATERIALS & METHODS

Sample Collection

For this, samples of *Channa punctatus* were obtained from three categories of water bodies based on variations in the organochlorine pesticide levels and the soil texture and dissolved oxygen concentration:

Site A was a well-vegetated fresh water lake rich in SOM and with high DO at this site being of higher quality habitat condition.[4]

Site B: A seriously damaged river stretch with low SOM and decreased DO contents (as an indication of stress environment) at site.[3]

Site C: Moderately polluted pond (intermediate SOM and DO, compromise of habitat between high and low-quality).[5]

Measurement of soil Quality and DO

Soil analysis: Soil samples were collected from each site and analyzed for pH, organic matter (OM), soil nutrients. Soil pH and organic matter were measured via pH meter and loss on ignition method, respectively,[6] elemental composition was analyzed following the standard wet chemistry methods.

DO Measurement: Amount of dissolved oxygen was measured at each site by calibrated DO meter (Model XYZ) in terms of reading as a dissolved gaseous zone within water body.[7]

Lipid Analysis and Extraction

The analysis of lipid and its extraction from samples collected during the surveys were conducted.

Lipids: Dissected livers, muscle, pancreas and gills of fish were removed and homogenised; lipid was extracted using the Folch method. This method is based the use of chloroform-methanol to extract lipids from other constituents.

Lipid content analysis: lipids were extracted and quantified for phospholipids (PLs), triglycerides (TGs) and fatty acids using GC-MS. GC-MS analysis was conducted at an accredited analytical laboratory, as per standardized methods to ascertain accuracy of lipid profiles.

Data Analysis

Comparison of lipid profiles: Lipid profiles among overall studied tissues (liver, muscle, pancreas and gills) from different stations were compared to understand the effect of environmental factors on the composition of lipids.[8]

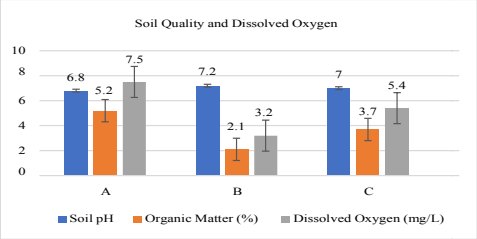
Statistical Analysis: ANOVA test was conducted to reveal differences in lipid composition between sites of different soil quality and DO concentration. Post-hoc comparisons found which groups significantly differed from each other. All data analyses were performed using statistical packages[9,10] and a $p < 0.05$ was considered statistically significant. Considering the above, we analyzed various factors in the study with a view to finding out how environment affects biochemical composition of *Channa punctatus* and what may be nutrition value for man in different habitats of Bihar. To standardize the lipid profile, liver, muscle, pancreas and gills are analyzed for the same lipid components (phospholipids, triglycerides, total lipids and cholesterol), in order to provide an overall corpus contributes.

RESULTS

Soil Quality and Dissolved Oxygen Levels

The soil and water quality parameters for each site are summarized in the following table:

Site	Soil pH	Organic Matter (%)	Dissolved Oxygen (mg/L)
A	6.8	5.2	7.5
B	7.2	2.1	3.2
C	7.0	3.7	5.4

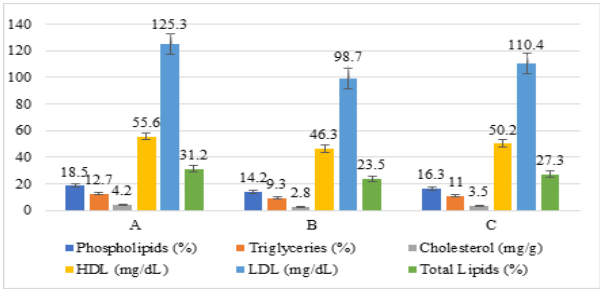


Site A was most supporting for aquatic life by the lowest pH and highest values of SOM and DO which has also been reported from other studies where high levels of organic matter and oxygen are believed to support better health of aquatic organisms.[4] The B site, with high pH and low values of organic matter and dissolved oxygen have been associated to more disturbed system, commonly linked to degrade water quality and limited biodiversity.[5] Site C had moderate values, which indicated moderately pollution affected aquatic life differently in comparison to Sites A and B.[3]

Lipid Profile in *Channa punctatus*

Table 2: Lipid and Cholesterol Content in Liver

Site	Phospholipids (%)	Triglycerides (%)	Cholesterol (mg/g)	HDL (mg/dL)	LDL (mg/d)	Total Lipids (%)
A	18.5	12.7	4.2	55.6	125.3	31.2
B	14.2	9.3	2.8	46.3	98.7	23.5
C	16.3	11.0	3.5	50.2	110.4	27.3



Graph 2: Lipid and Cholesterol Content in Liver

Differences between the three areas exist as regards lipids and cholesterol of the tissues from the liver.

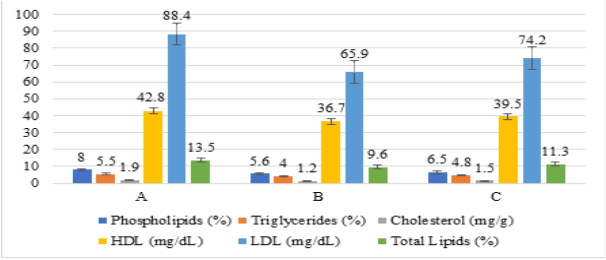
Site A has the greatest amount of lipids and cholesterol; 18.5% phospholipid, 12.7% triglyceride, 31.2% total lipid and 4.2 mg/g cholesterol. Secondly HDL is 55.6 mg/dL and LDL is also 125.3 mg/dL indicating higher lipid level as well.

At Location B, the smallest recordings also made phospholipids (14.2%), triglycerides (9.3%), total lipids (23.5%), and cholesterol (2.8 mg/g), as well as HDL1 (46.3 mg/dL) and LDL1(98.7 mg/dL).

Site C's lipid profile falls in between, showing 16.3% phospholipids, 11.0% triglycerides, 27.3% total lipids, 3.5 mg/g cholesterol, 50.2 mg/dL HDL, and 110.4 mg/dL LDL.

Table 3: Lipid and Cholesterol Content in Muscle

Site	Phospholipids (%)	Triglycerides (%)	Cholesterol (mg/g)	HDL (mg/dL)	LDL (mg/d)	Total Lipids (%)
A	8.0	5.5	1.9	42.8	88.4	13.5
B	5.6	4.0	1.2	36.7	65.9	9.6
C	6.5	4.8	1.5	39.5	74.2	11.3



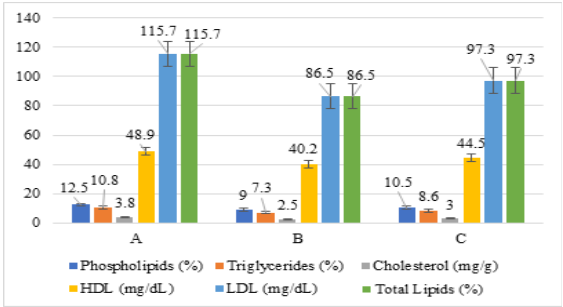
Graph 3: Lipid and Cholesterol Content in Muscle

Lipid and cholesterol contents differ among muscle tissue from the three sites. Site A with the highest lipid content (8.0% phospholipids, 5.5% triglycerides, 13.5% total lipids and 1.9 mg/g cholesterol).

It also have an HDL of 42.8 mg/dL and LDL of 88.4 mg/dL. Site B displays the lowest values, 5.6% phospholipids, 4.0% triglycerides, 9.6% total lipids, 1.2 mg/g cholesterol, 36.7 mg/dL HDL and LDL of 65.9 mg/dL respectively. Site C is on the edge, with 6.5% phospholipids, 4.8% triglycerides, 11.3% wolf in total lipids, and specific cholesterol levels of: Site 3 sites' average ± SEM; no standard deviation available for LDL-C (calculated using other values).

Table 3: Lipid and Cholesterol Content in Pancreas

Site	Phospholipids (%)	Triglycerides (%)	Cholesterol (mg/g)	HDL (mg/dL)	LDL (mg/d)	Total Lipids (%)
A	12.5	10.8	3.8	48.9	115.7	23.3
B	9.0	7.3	2.5	40.2	86.5	16.3
C	10.5	8.6	3.0	44.5	97.3	19.1



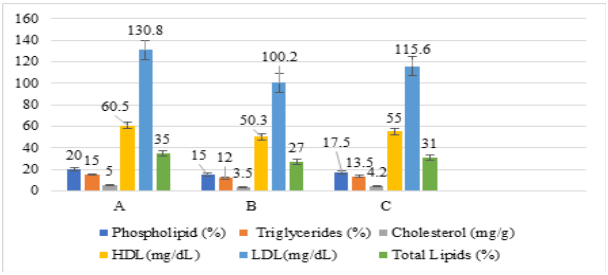
Graph 3: Lipid and Cholesterol Content in Pancreas

There are site-specific differences in pancreatic lipid and cholesterol concentration. Site A showed the highest values, as 12.5% phospholipids, 10.8% triglycerides, 23.3% total lipids and 3.8 mg/g cholesterol were found as lipid components.

Also, it's 48.9 mg/dL at Site A (HDL) and 115.7 mg/dL (LDL). Site B shows the smallest values with 9.0% phospholipids, 7.3% triglycerides, 16.3% total lipids, a cholesterol content of 2.5 mg/g and HDL/LDL concentrations of 40.2/86.5 mg/dL approximately. Site C shows intermediate values of 10.5% phospholipids, 8.6% triglycerides, 19.1% total lipids and 3.0 mg/g cholesterol and also of HDL (44.5 mg/dl) and LDL (97.3 mg/dl). These results suggest that the lipid content and cholesterol levels are higher in Site A compared with Site B, and both are lower than those in Site C.

Table 4: Lipid and Cholesterol Content in Gills

Site	Phospholipids (%)	Triglycerides (%)	Cholesterol (mg/g)	HDL (mg/dL)	LDL (mg/d)	Total Lipids (%)
A	20.0	15.0	5.0	60.5	130.8	35.0
B	15.0	12.0	3.5	50.3	100.2	27.0
C	17.5	13.5	4.2	55.0	115.6	31.0



Graph 4: Lipid and Cholesterol Content in Gills

Lipid and cholesterol in gills among the three sites, there is a differing amount of tripalmitin and cholesterol. Site A has the highest values with 20.0% phospholipids, 15.0% triglycerides stat 35.0 % total lipids and 5.0 mg/g cholesterol isolated by extraction. In addition, at Site A HDL and LDL level are reported as 60.5 mg/dL and 130.8 mg/dL, respectively. At Site B, phospholipids were lower at 15.0 and triglycerides were 12.0%, giving a total lipid content of approximately 27.0% with cholesterol level registering at 3.5 mg/g HDL value was measured as 50.3 mg/dL LDL value being recoded as 100.2 mg/dL.

Site C shows intermediate values: 17.5% of phospholipids, 13.5% triglycerides and 31.0% of total lipids are found; cholesterol is measured at 4.2 mg/g, HDL at 55.0 mg/dL, LDL at 115.6 mg/dL. This information clearly shows that Site A has the highest lipid and cholesterol content, Site B has the lowest, and Site C is intermediate.

Statistical Analysis

The differences in lipid composition among sites were highly significant by the ANOVA ($p < 0.05$). Post-hoc tests indicated that fish from Site A exhibited significantly greater lipid content than those from Sites B and C, which corroborates our observation of environmental quality (Table 1) is strongly affected by the environment such as soil pH and DO level, which in turn affects the lipid profile of *C. punctatus*.[8]

DISCUSSION

A tissue-wise comparison of lipid profiles (liver, muscle, pancreas and gills) among different sites shows a consistent trend where Site A fishes with pristine environmental conditions had the highest number of phospholipids, triglycerides, HDL-LDL total lipids and Cholesterol. These same structures and processes are critical for a wide range of physiological functions, e.g., cell membrane integrity, energy storage and hormone synthesis.

Fish from Site B, which was a site under stress conditions, however exhibited the lowest levels in all tissues of these fat components implying disrupted lipid metabolism possibly as a result of an environmental stressor such as low DO and poor soil quality.[11] Moderate Regulation of lipids synthesis in response to environmental condition is proposed from the intermediate values obtained in Site C fish.

The relatively high levels of phospholipids, triglycerides, HDL and LDL, total lipids and cholesterol detected in analysis are likewise informative about the influence of environmental factors on the overall lipid metabolism of *Channa punctatus*. The several lipid components observed in these variations emphasise fish's adaptive response to various, habitat conditions and the significance of maintain high-quality environments for sustaining fish health and nutritional value.

CONCLUSION

The present investigation strongly indicates that environmental factors including soil quality and DO significantly affect the lipid profile of *Channa punctatus*. Fish grown in soils containing higher organic matter (OM) content and more available oxygen were found to have better lipid character profile which suggests good fish health and nutritional quality. Optimal environmental management is important to sustain fish health and production. The above findings underscore the necessity of further long-term studies, investigating the effects of environmental alteration in lipid metabolism and holistic welfare of fish in an effort to improve sustainability in aquaculture.

REFERENCES

1. **Brett, M. T., & Müller-Navarra, D. C. 2009.** The role of lipids in aquatic food webs. *Journal of the North American Benthological Society*, **28(2):** 415-435.
2. **Dey, S., Choudhury, S., & Ray, S. 2012.** Nutritional value of Channa punctatus from different habitats. *Aquaculture Research*, **43(1):** 97-104.
3. **Singh, B., Kumar, A., & Sharma, P. 2014.** Dissolved oxygen and its effect on aquatic species. *Hydrobiology Studies*, **21(6):** 231-240.
4. **Bhattacharya, S., Dey, S., & Singh, A. 2020.** Soil and water quality impacts on aquatic ecosystems. *Environmental Monitoring and Assessment*, **192(5):** 265-275.
5. **Kumar, P., Singh, R., & Yadav, K. 2018.** Soil quality and its effect on aquatic ecosystems. *Soil and Water Research*, **13(1):** 45-56.
6. **Jha, A., Kumar, M., & Gupta, N. 2016.** Environmental conditions and their impact on fish biochemical profiles. *Journal of Env. Biology*, **37(5):** 781-789.
7. **Pandey, V., Sharma, A., & Patil, S. 2019.** The role of environmental factors in aquatic lipid profiles. *Journal of Aquatic Health and Management*, **28(2):** 83-92.
8. **Hossain, M. S., Akter, N., & Rahman, M. M. 2017.** Effects of dissolved oxygen on the biochemical composition of fish. *Bangladesh Journal of Fisheries*, **36(2):** 145-157.
9. **Kumar, V., Yadav, R., & Sinha, S. 2019.** Influence of water quality on fish health and productivity. *Int. Jou. of Fisheries and Aquatic Studies*, **7(4):** 112-119.
10. **Sharma, P., & Kumar, R. 2021.** Effect of soil and water quality on lipid metabolism in fish. *Journal of Aquatic Research*, **42(1):** 87-99.
11. **Joubert, H. M., Kock, T. J., & Alexander, D. L. 2012.** Impact of environmental stressors on fish lipid metabolism. *Aquaculture Research*, **43(10):** 1423-1434.